

Coarse Capacitance (Cc) Characterization Flow

1. Testbench: PrimeSim_default_justCC

Design: Tese/Cc_Characterization/schematic_justCC

Analysis: AC with two Points of Interest (POI): 6 GHz ($f_{\min}$) and 9.5 GHz ($f_{\max}$).

Sweep: Temperature sweep (temp) from -40°C to 125°C (linear step 16.5).

Corners: Typ25, bcQ-40, wcQ125 (Combinations of Temperature and RC Extraction files).

1.1 Minimum Frequency ($f_{\min} = 6\text{ GHz}$) / $C_{\max}$ Analysis

OUTPUT	EXPRESSION
I_{dev} (Device_current)	i(/V45/PLUS)
$C_{\text{max,meas}}$	abs(imag(Device_current) / (2 * 3.14159 * 6e9))
$F_{\min}$	1 / (2 * 3.1416 * sqrt(236e-12 * Cap_measured_max))

1.2 Maximum Frequency ($f_{\max} = 9.5\text{ GHz}$) / $C_{\min}$ Analysis

OUTPUT	EXPRESSION
I_{dev} (Device_current)	i(/V45/PLUS)
$C_{\min,meas}$	abs(imag(Device_current) / (2 * 3.14159 * 9.5e9))
$F_{\max}$	1 / (2 * 3.1416 * sqrt(236e-12 * Cap_measured_min))